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Codominant inheritance of polymorphic color mutant and characterization of a bisexual mutant of *Gracilaria caudata* (Gracilariales, Rhodophyta)

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Abstract

The red alga *Gracilaria caudata* is one of the main agarophytes exploited on the Brazilian coast. A greenish-brown mutant was found growing side-by-side with the common red wild type in a natural population. In this work, we tested the hypotheses that the greenish-brown phenotype has a recessive nuclear inheritance and less fitness (growth, photosynthesis, number of apices, secondary attachment percentage, and number of cystocarps) under laboratory conditions when compared with the red phenotype. Crosses between plants of the same color (red or greenish-brown) showed tetrasporophytic and subsequent gametophytic generations with parental phenotype. However, heterozygous tetrasporophytes, obtained from crosses between plants with different colors, showed a brown phenotype. From their progeny, we concluded that the brown phenotype is due to a monolocus Mendelian genetic system with two codominant alleles. Bisexual gametophytes were found in the progeny of tetrasporophytic color mutants, which is an unusual and poorly known characteristic in the life cycle of Gracilariales. Red female gametophytes (*rd* genotype) presented better performance in relation to the greenish-brown female gametophytes (*gb* genotype): growth rates ($rd = 17.14 \pm 1.36$, $gb = 13.81 \pm 0.95$); number of apices ($rd = 41.05 \pm 6.35$, $gb = 16.00 \pm 5.47$); secondary attachment percentage ($rd = 68.75 \pm 23.93$, $gb = 25 \pm 0.00$); and number of cystocarps ($rd = 38.61$, $gb = 18.81$). Our results explain the maintenance of the greenish-brown mutant and the predominance of red specimens in natural population. The life cycle was completed in 5 months, one of the shortest times reported for *Gracilaria*, highlighting the importance of the species as a possible model organism for reproductive studies and in selecting strains for use in mariculture.

Keywords *Gracilaria caudata* · Color inheritance · Bisexuality · Secondary attachment

Introduction

The red algal Order Gracilariales has been the target of many studies because it includes species that constitute the main world source of agar (Porse and Rudolf 2017). In Brazil, the exploitation of natural populations for the production of this phycocolloid began in the 1960s (Câmara-Neto 1987) and despite attempts, no commercial crops can be found. Currently, the exploitation of this resource occurs mainly in the northeastern region by artisans using initial biomass or seedlings from natural beds (Miranda et al. 2012; Hayashi et al. 2014; Simioni et al. 2019). This activity has led to the depletion of many natural beds along the Brazilian coast, as already observed for other regions of the world (McHugh 2003; Hayashi et al. 2014). In this sense, the selection of more productive strains can be considered a viable alternative to satisfy demand and preserve natural populations. *Gracilaria* has species that exhibit great phenotypic plasticity. Furthermore, the discovery of cryptic species through molecular techniques (Cohen et al. 2004; Guillemin et al. 2008; Destombe et al. 2010; Lyra et al. 2016) has made its taxonomy a subject of great debate (see Gurgel et al. 2018; Iha et al. 2018). Therefore, we chose in this work to refer *Gracilaria caudata* rather than *Crassiphycus caudatus* as proposed by Gurgel et al. (2018) and Guiry et al. (2018).

Like most Florideophyceae, *Gracilaria* presents a triphasic life cycle ("Polysiphonia-type") characterized by the alternation of diploid (tetrasporophyte and carposporophyte) and haploid (dioecious gametophyte) phases. Tetrasporophytes and gametophytes are isomorphic and free-living individuals, while the carposporophyte is dependent on the female gametophyte. After meiosis, tetraspores are released, and following

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their germination, the proportion between female and male gametophytes has been reported as 1:1 (Kain and Destombe 1995). Usually, *Gracilaria* species have a reddish color owing to the presence of phycobiliproteins, mainly phycoerythrin, masking chlorophyll *a* and carotenoids. However, color mutants can be found in natural populations (van der Meer 1990). In the genus, the first genotypic color mutant was reported for *G. tikvahiae*, which emerged spontaneously in the laboratory (as *Gracilaria* sp., van der Meer and Bird 1977). In natural populations, they were found for *G. conferta* and *G. verrucosa* (Levy and Friedlander 1990), *G. chilensis* (Levy and Friedlander 1990), *G. domingensis* (Plastino et al. 1999), *G. birdiae* (Costa and Plastino 2001; Plastino et al. 2004), *G. cornea* (Ferreira et al. 2006), *G. caudata* (Faria and Plastino 2016), and *G. debilis* and *G. edulis* (Veeragurunathan et al. 2016). Color mutants also have been described for other genera of Rhodophyta (van der Meer 1990, for a review). However, most knowledge about this subject comes from the studies on color mutants of *G. tikvahiae* obtained by mutagenesis (van der Meer 1990).

Color mutants have been used in genetic studies since they contribute to the detection of fertilization, distinction between sexual and asexual processes (van der Meer et al. 1984), and between self-fertilization and cross-fertilization in monoecious species (van der Meer 1987). In addition, the use of these mutants has revealed in situ tetraspore germination (Plastino et al. 1999), failures during the cytokinesis of tetrasporangia (van der Meer 1977), and sexual maturation of microscopic female gametophytes (van der Meer and Tood 1977). However, before using color mutants as a genetic marking tool, it is necessary to understand if these phenotypes are reversible or not (stability), as well as its inheritance pattern (van der Meer 1990). In most studies in which the mutant color is genetically inherited, it has a nuclear inheritance either recessive (van der Meer 1990 for a review), dominant (van der Meer 1979a) or codominant (van der Meer 1979b; Plastino et al. 1999). In other cases, the mutations are cytoplasmic (van der Meer 1978; Zhang and van der Meer 1988; Costa and Plastino 2011). The study of these mutants provides a better understanding of sexual reproduction and genetics in these species with a haploid-diploid life cycle (van der Meer and Tood 1977). This knowledge could be used in the context of varietal selection of more productive strains for mariculture, targeting phycocolloids, pigments and other products. This would require an understanding of the physiology of such mutants since many characteristics are positively or negatively correlated with others, and consequently, they can affect the choice of strategies to be used in breeding programs (Robinson et al. 2013).

Gracilaria species expresses sex in the haploid phase differently of most animals and land plants in which sex is expressed in the diploid phase (males are heterogametic XY or ZW, whereas females are homogametic XX or ZZ). This genetic mechanism of sex determination is termed UV system and is mediated by sex-determining regions (SDRs) present in the female (U) and male (V) sex chromosomes (see Coelho et al. 2018 for a review). The diploid phase is always heterogametic (UV) and, after meiosis, it is expected the segregation of these two sex chromosomes in 1:1. Based on test crosses under laboratorial condition, van der Meer and Tood (1977) studied sexual differentiation in *G. tikvahiae*. The authors suggested that the 1:1 sex ratio between female and male gametophytes was due to the Mendelian segregation of a pair of alleles (mt^{m} for males and mt^{f} for females). Moreover, they suggested that such alleles could recombine mitotically in the tetrasporophytic phase, resulting in a diploid tissue with male or female reproductive characteristics. Later studies in *G. tikvahiae* showed the occurrence of bisexual gametophytes (monoicous). Under laboratory conditions, many crosses were performed and the phenomenon was explained by the interaction of two alleles located on different loci: a recessive allele called *bi* would act on the mt^{m} allele allowing the expression of female characteristics (fertilization and subsequent presence of cystocarps) in male gametophytes (van der Meer et al. 1984; 1986). In *Gracilaria* there are few studies that have characterized molecular markers linked to sex determination. Some of them were performed for *G. gracilis* (Martinez et al. 1999), *G. changii* (Sim et al. 2007), and *G. chilensis* (Guillemin et al. 2012). However, only Martinez et al. (1999) and Guillemin et al. (2012) found a characteristic Mendelian inheritance of the sexual markers. The UV system is also characteristic of some brown algae with haploid-diploid life cycle as *Ectocarpus* sp. Using molecular techniques, this species has been the target of several studies related to sex determination and theoretical predictions about the evolution of haploid UV systems, being one of the few species that have a wide characterization of the SDRs from U and V chromosomes (Ahmed et al. 2014; Coelho et al. 2019).

Gracilaria caudata is widely distributed along the Brazilian coast. Common on bedrock, it occurs mostly in protected bays and turbid waters, from the intertidal zone to the subtidal fringe (Plastino and Oliveira 1997). Populations of *G. caudata* have been exploited since the 1970s for the production of agar destined for the domestic and food market (Oliveira and Miranda 1998). Currently, it is one of the main exploited species for agar production in the country (Torres et al. 2019). Based on its economic importance, several studies have been carried out, and some have demonstrated better growth rates at 18–30 °C and salinity tolerance between 10 and 60 psu (as *Gracilaria* sp., Yokoya and Oliveira 1992a, 1992b, or as *Hydropuntia caudata*, Miranda et al. 2012). Both ecotypes (Araújo et al. 2014; Faria et al. 2017) and high genetic diversity of *G. caudata* were observed along the Brazilian coast (Ayres-Ostrock et al. 2016, 2019; Chiaramonte et al. 2019).

Recently, a greenish-brown mutant of *G. caudata* was found growing side-by-side with red wild-type individuals (most common) in a natural population of the northeastern coast of Brazil (Faria and Plastino 2016). Considering the importance of this Brazilian resource, we tested the hypotheses that (i) the greenish-brown phenotype has a recessive nuclear inheritance in relation to red, in which heterozygous tetrasporophytes, obtained after cross tests between gametophytes with different phenotypes, would have red phenotype and the *rdgb* and *gbrd* genotypes. In the next gametophytic generation specimens with red and greenish-brown phenotypes would be observed in the ratio of 1:1; and (ii) specimens with greenish-brown phenotype would have less fitness (growth, photosynthesis, number of apices, secondary attachment percentage, and number of cystocarps) under laboratory conditions compared with the red phenotype. We also characterized bisexual gametophytes found in the progeny of tetrasporophytic color mutants, which is an unusual phenomenon in the life cycle of Gracilariales. It is expected that this study will contribute to the interpretation of possible adaptive advantages in maintaining this mutant in nature, as well as provide input for its use in genetic studies and mariculture.

Materials and methods

Biological material

Red (wild-type) and greenish-brown cystocarpic plants of *Gracilaria caudata* were collected at Paracuru Beach (3.4° S 39.07° W), Ceará State, Brazil, in September 2012. Voucher specimens are deposited in the herbarium of the University of São Paulo (SPF 57843 and SPF 57848). In laboratory conditions, carpospores released from these plants were cultivated, and unialgal non-axenic cultures were established (Plastino and Oliveira 1990). Tetrasporophytes derived from carpospores were kept in the Gracilariaceae Germplasm Bank of the University of São Paulo (Costa et al. 2012). For our purposes, tetrasporophytes with different phenotypes (red and greenish-brown) were cultivated until the liberation of tetraspores and the obtaining of plantlets.

General culture conditions

Plants were cultivated in von Stosch enriched seawater with modifications (Ursi and Plastino 2001) and diluted to 25% with sterile seawater (salinity at 32 psu). They were kept in a temperature-controlled room at 25 ± 1 °C with a photoperiod of 14 h. Photosynthetically active radiation (PAR) was 70 μmol photons m^{-2} s^{-1} provided by Osram 40 W daylight fluorescent tubes, and PAR was measured with a quantameter (Li-COR model L1–185). Cultures were aerated for 30 min h^{-1}, and the medium was renewed weekly.

Color inheritance

Cultures of red and greenish-brown gametophytes of *G. caudata* (*rd* and *gb* genotypes) were established from tetraspores released by red and greenish-brown tetrasporophytes, respectively. To determine the color inheritance pattern, we performed crosses (Fig. 1) between female and male individuals of the same phenotype (♀*rd* x ♂*rd* and ♀*gb* x ♂ *gb* genotypes) and between different phenotypes (♀*rd* x ♂*gb* and ♀*gb* x ♂*rd* genotypes). Before starting crosses, apices cut from female gametophytes were kept isolated from males for at least 2 months to ensure the absence of fertilized carpogonia and during the experimental period, female branches of each color were kept isolated from male branches to check for the existence of monoecious or apomictic plants. For the crosses, an apical female branch and a male branch were incubated together for 48 h in 500 mL Erlenmeyer flasks, containing 450 mL of enriched seawater under general culture

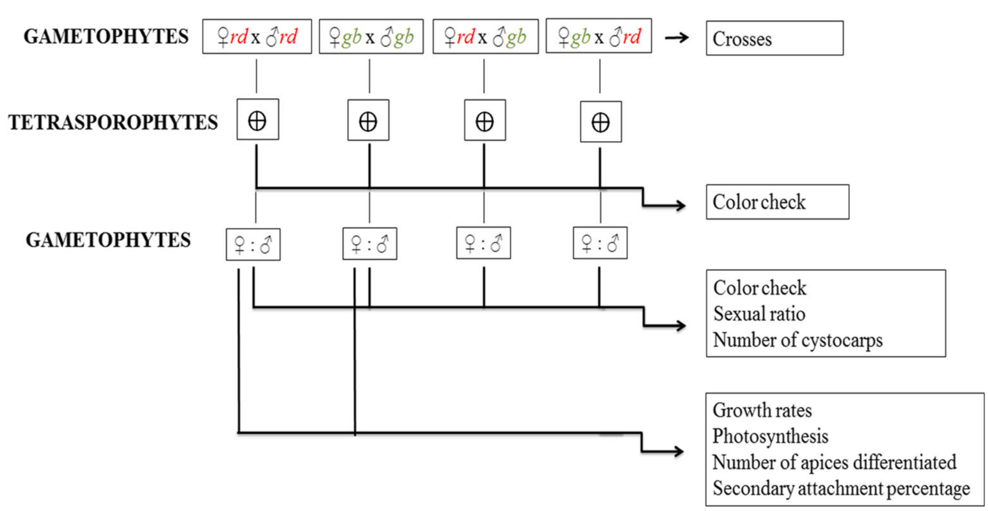

Fig. 1 Flow chart of the steps to obtain *rd* (red) and *gb* (greenish-brown) gametophytes of *Gracilaria caudata* to be used in experiments of color inheritance and evaluations of growth and photosynthesis rates, attachment percentage, number of apices and, number of cystocarps

conditions. After this period, the male branch was then removed and the female cultivated until producing cystocarps and releasing carpospores. We did this procedure only once per combination of genotypes. Ten carpospores of each resulting genotype (*rdrd*, *gbgb*, *rdgb*, and *gbrd*—the first written "allele" refers to the genotype of the female gametophyte that gave rise to these carpospores, while the second indicates the male genotype) from the crosses were collected and grown under the general culture conditions in order to obtain fertile tetrasporophytes, as well as verify color phenotypic ratios. We randomly select one fertile tetrasporophyte of each genotype (*rdrd*, *gbgb*, *rdgb*, and *gbrd*), and from each of them we collected 25 tetraspores, which were cultivated in a group of five per flask. They originated gametophytes and, when they were 84 days old, the sex and color phenotypic ratios were analyzed. At the same time, the number of cystocarps was evaluated under stereomicroscope (Leica WILD M3C 10x). The mean number of cystocarps by female gametophytes was calculated by the number of cystocarps present in all female gametophytic branches divided by the number of female gametophytes by flask. When bisexual phenotypic gametophytes (*bi* genotype) were found, we performed a viability analysis of the "carpospores" released from their cystocarps. For this, we selected a single cystocarp from each *bi* gametophyte. After that, we selected twenty carpospores and cultivated them for a period of 60 days.

Physiological analyses

Red and greenish-brown female gametophytes originated from the germination of tetraspores released by red (*rdrd* genotype) and greenish-brown (*gbgb* genotype) tetrasporophytes, respectively, were used in these experiments. Each replicate ($n = 4$) was composed of four apical segments 1 cm in length, not branched, and coming from a unfertilized female gametophyte selected under stereomicroscopy. The apical segments were then cultivated in 1-L Erlenmeyer flasks. However, the volume of enriched seawater used varied according to the experiment (850 or 950 mL), as well as the experimental period (14 or 21 days). Further modifications of culture conditions will be mentioned when necessary. Such gametophytes were analyzed for growth rates and in vivo chlorophyll fluorescence, as well as the number of apices differentiated and the thallus attachment percentage on the culture flask (Fig. 1).

Growth rates and in vivo chlorophyll fluorescence

The plants were cultivated in Erlenmeyer flasks containing 850 mL of enriched seawater under general conditions ($n = 4$). The growth rates were analyzed after 14 days of cultivation. They were assessed by measurements of fresh mass and estimated by average of the following equation: $GR = [(M_t/M_0)^{1/t} - 1] \times 100$, where M_t is the final fresh mass, M_0 is the initial fresh mass, and t is the time (Yong et al. 2013).

In vivo chlorophyll fluorescence was measured after 14 days of cultivation using a Diving-PAM fluorometer (Walz, Germany). The measurements were done after 4–7 h of exposure to light in the culture chamber. Apical segments were arranged on a magnetic sample holder to avoid overlap. We used 11 levels of irradiance to construct electron transport rate (ETR) × irradiance (PAR) curves: 6, 19, 41, 75, 113, 156, 233, 323, 497, 738, and 1113 μmol photons m^{-2} s^{-1}. The apical segments were exposed for 20 s at each irradiance, spaced with saturation pulse of 0.8 s, approximately 6000 μmol photons m^{-2} s^{-1}. The effective quantum yield ($\Delta F/F_m'$) was provided by the Diving-PAM after an initial saturation pulse when the samples were at a very low-intensity pulse of blue light (approximately 0 μmol photons m^{-2} s^{-1}). The following parameters were calculated (Platt et al. 1980) and analyzed (Schreiber 2004): photosynthetic efficiency (α_{ETR}), maximum electron transport rate (ETR_{max}), light saturation (Ek_{ETR}), and effective quantum yield ($\Delta F/F_m'$).

Number of apices

A new cultivation was established from apical segments of *rd* and *gb* female gametophytes (1 cm in length, not branched). The number of apices differentiated in each apical segment was analyzed weekly over a period of 21 days. The algae were cultivated in Erlenmeyer flasks containing 950 mL of enriched seawater under general conditions ($n = 4$).

Attachment percentage

We analyze the attachment percentage to the culture flask by counting the number of secondary attachment disc with the naked eyes. For this experiment, a new cultivation from apical segments of *rd* and *gb* female gametophytes (four apices per flask $n = 4$) was also established under general conditions with the following modifications: irradiance of 25 ± 5 μmol photons m^{-2} s^{-1} and 950 mL of enriched seawater without aeration and exchange of seawater. After a period of 14 days we obtained the number of secondary attachment discs and analyzed attachment percentage by the following equation for each replicate ($n = 4$): $A\% = A/T * 100$, where A is the total of apices attached to the culture flask, and T is the total of apices present in the culture flask.

Statistical analysis

The normality and homogeneity assumptions of variances were tested using the Kolmogorov-Smirnov and Cochran tests, respectively. When necessary, logarithmic [$x = \log(x + 1)$] and arcosine (arcosine (root) (value)) transformations were employed and retested (Zar 1996). Data, such as growth rate, photosynthetic parameters (EK_{ETR}, $\Delta F/F_m'$, α_{ETR} and ETR_{max}), number of cystocarps differentiated, and attachment percentage were analyzed by one-way ANOVA, while the number of apices differentiated were subjected to repeat measures analyses of variance (RMANOVA). In all cases, the posteriori Newman-Keuls test was used to establish statistical differences. Statistical analyses were done using the Statistica 10 program, considering $p < 0.05$ as significant. The sex and color ratios among gametophytes derived from *rdrd*, *gbgb*, *rdgb*, and *gbrd* tetrasporophytes were determined using the chi-square test.

Results

The required period to complete the life cycle of *G. caudata* (since the crossing until F2 fertile gametophytic generation) varied according to the type of initial crossing performed: ♀*rd* × ♂*rd* (157 days); ♀*gb* × ♂*gb* (145 days); ♀*rd* × ♂*gb* (138 days); and ♀*gb* × ♂*rd* (132 days).

Color inheritance

All crosses (♀*rd* × ♂*rd*, ♀*gb* × ♂*gb*, ♀*rd* × ♂*gb*, and ♀*gb* × ♂*rd*) were positive, and no cystocarps were formed in the absence of males. Carposporophytes resulting from the different crosses produced dark red spore masses inside red pericarps and released reddish carpospores. Cystocarp differentiation, with the naked eye, and release of carpospores were respectively observed after 6 and 18 days of contact between female and male gametophytes, regardless of the cross.

Carpospores resulting from all crosses developed into small basal disks and soon gave rise to red erect plantlets. Differences in color appeared in individuals about 1 cm in length and approximately 45 days old. Tetrasporophytes derived from ♀*rd* × ♂*rd* presented red phenotype (*rdrd* genotype), while tetrasporophytes derived from ♀*gb* × ♂*gb* presented a greenish-brown phenotype (*gbgb* genotype). In addition, a brown phenotype was observed in *rdgb* and *gbrd* genotypic plants, but the color differentiation between them and the other tetrasporophytes was obviously visible only when those strains reached about 4–5 cm in length and already had tetrasporangia.

Tetraspores obtained from *rdrd*, *gbgb*, *rdgb*, and *gbrd* tetrasporophytes developed small basal disks and soon gave rise to red erect plantlets, which were kept in culture until complete at 106 days old. Differences in sex, color and morphology were identified in the next gametophytic generation, depending on parental tetrasporophyte genotype.

Plantlets obtained from *rdrd* tetrasporophytes produced female and male red gametophytes in a ratio of 1:1 (14♀: 11♂, $X^2 = 0.36$), (Table 1, Figs. 2–3). Red cystocarps were observed when female gametophytes were 62 days old. By that time, spermatangia on male gametophytes had already been observed. These cystocarps were initially visualized as superficial red spots, which later acquired a protruding shape from the thallus and released carpospores (Fig. 2b-c).

Plantlets obtained from *gbgb* tetrasporophytes gave rise to greenish-brown female and bisexual gametophytes (*bi*) in a ratio of 1:1 (11♀:14*bi*, $X^2 = 0.36$) (Table 1, Fig. 3). At 56 days of age, we observed the beginning of cystocarp formation in female gametophytes. Cystocarps (pericarp and mass of carpospores) were red, as well as the portions of the thallus adjacent to them (Fig. 2b and c), while male gametophytes were identified by the presence of a full spermatangia thallus. After 20 days, however, these male individuals (76 days old) started to produce cystocarps along the thallus, which were bordered by spermatangia. At 99 days of age, all male gametophytes had developed at least one cystocarp; thus, we could call them as bisexual (*bi*) (Fig. 4). Initially, their cystocarps had a concave shape, which later protruded to the thallus and released red carpospores 27 days after the beginning of cystocarp formation.

Plantlets obtained from brown tetrasporophytes *rdgb* and *gbrd* gave rise to three reproductive classes: female, male, and bisexual gametophytes (11♀: 08♂: 06*bi* and 9♀: 11♂: 05*bi*, respectively) with red or greenish-brown color. Gametophytes showed a 1:1 color ratio (*rd*:*gb*), regardless of the tetrasporophyte they came from *rdgb* or *gbrd* (Table 1; Fig. 3). Similarities were also observed in relation to the development of cystocarpic plants at 49 days of age, regardless

Table 1 *Gracilaria caudata*—sex and color phenotypic ratios of gametophytes derived from the germination of tetraspores released by *rdrd*, *gbgb*, *rdgb*, and *gbrd* genotypic tetrasporophytes

Crosses ♀ × ♂	Tetrasporophytes (Thallus color and genotype)	Gametophytes (Sex—♀: ♂:*bi*)	Gametophytes (Number and thallus color)
rd × *rd*	Red—*rdrd*	14:11:0	25 red
gb × *gb*	Greenish-brown—*gbgb*	11: 0: 14	25 greenish-brown
rd × *gb*	Brown—*rdgb*	11: 8: 6	12 red, 13 greenish-brown
gb × *rd*	Brown—*gbrd*	9: 11: 5	15 red, 10 greenish-brown

The first column presented the crosses between red and greenish-brown gametophytes (*rd* and *gb* genotypes, respectively)

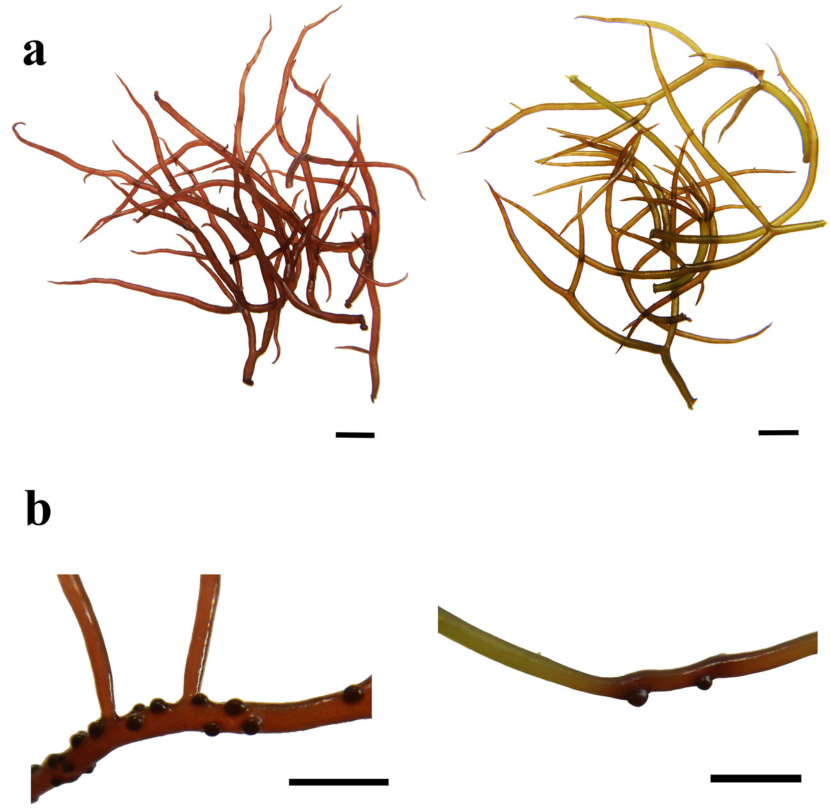

Fig. 2 *Gracilaria caudata*—(a) unfertilized apices of red (left) and greenish-brown (right) female gametophytes obtained from the germination of tetraspores released by red and greenish-brown tetrasporophytes, respectively. (b) Branches of red (left) and greenish-brown (right) cystocarpic plants. Scale: 5 mm

of tetrasporophyte origin. However, bisexual gametophytes differed from each other in relation to the time required to initiate the development of cystocarps: 71 days for those coming from *rdgb* tetrasporophyte and 65 days for those coming from *gbrd* tetrasporophyte. At 106 days old, *bi* gametophytes from *rdgb* and *gbrd* tetrasporophytes had, on average, 2.8 and 2.6 cystocarps per individual, respectively. The release of carpospores from these *bi* gametophytes occurred 21 and 19 days after cystocarp observation, respectively.

Carpospores released by *bi* gametophytes developed into plantlets. All plantlets derived from a single cystocarp presented similar reproductive structures: from some cystocarps, all plantlets showed tetrasporangia while from others, all plantlets showed spermatangia. However, they varied according to the parental *bi* gametophyte from which they came (Table 2). Plantlets obtained from *bi* 03, *bi* 04, *bi* 05, and *bi* 08 (Table 2) developed spermatangia and 22 days after the beginning of the cultivation, a single specimen obtained from *bi* 08 began to produce cystocarps along the thallus. All plantlets obtained from *bi* 02, *bi* 06, and *bi* 07 developed tetrasporangia. Nonetheless, at 22 days from the beginning of cultivation, we observed in a single specimen from *bi* 07 the formation of a cystocarp in addition to spermatangia.

To better understand the inheritance of bisexuality we selected three apices 2 cm in length from the same greenish-brown (*gb* genotype) and red (*rd* genotype) male gametophytes used in the initial cross tests. These three apices were kept together in Erlenmeyer flasks containing 850 mL of enriched seawater under general culture conditions for a month. As a result, we observed that no *rd* genotyped male gametophytes produced cystocarps, while all "*gb* genotyped male gametophytes" had differentiated at least one cystocarp, evidencing that they were also bisexual.

Differences in the number of cystocarps were observed among female fertilized gametophytes originated from different genotypic tetrasporophytes (*rdrd*, *gbgb*, *rdgb*, and *gbrd*) at 84 days of age. The largest median number of cystocarps (38.61) was observed for female gametophytes from *rdrd* genotypic tetrasporophytes ($F = 5.38$, $p < 0.01$), when compared with the others (from *gbgb* = 18.81; *rdgb* = 18.50; and *gbrd* = 19.25), which had a similar number of cystocarps among them ($p > 0.94$).

Growth rates

Red female gametophytes of *G. caudata* showed higher growth rates (17.14 ± 1.36) when compared with greenish-brown gametophytes (13.81 ± 0.95) after 14 days of cultivation in the same conditions ($F = 15.95$, $p < 0.05$) (Fig. 5a).

In vivo chlorophyll fluorescence

Greenish-brown female gametophytes showed higher values of ETR_{max} ($rd = 1.97 \pm 0.06$ and $gb = 2.33 \pm 0.09$; $F = 41.12$,

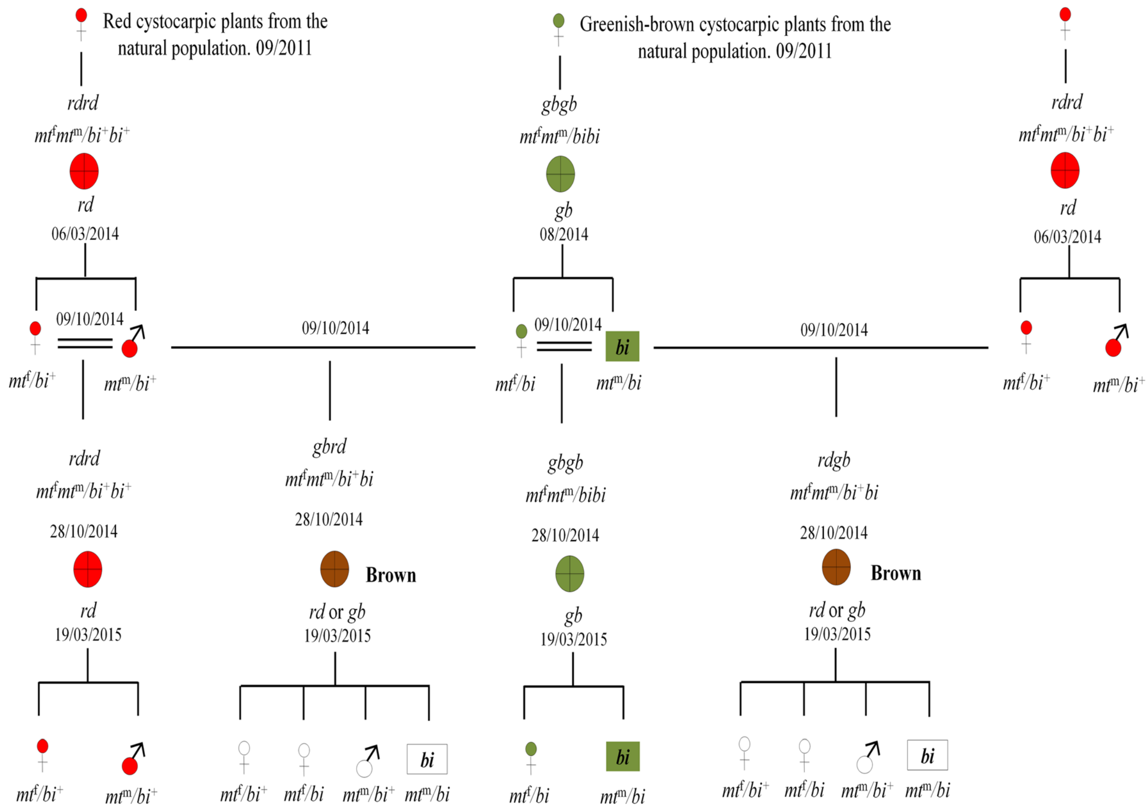

Fig. 3 Genealogy of *Gracilaria caudata* obtained of red and greenish-brown cystocarpic plants from a natural population of Praia da Pedra Rachada Beach – Ceará, Brazil. Abbreviations: ⊕ tetrasporophyte; ♀, female gametophyte; ♂, male gametophyte; *bi*, bisexual gametophyte; *rd*, red; *gb*, greenish-brown; *rdrd,* red; *gbgb*, greenish-brown; *gbrd* and *rdgb*, brown; —, crossing; ═, crossing between gametophytes obtained from the germination of tetraspores released by the same tetrasporophyte; mt^f, allele that determines the expression of female characteristics (carpogonia production and development of cystocarps); mt^m, allele that determines the expression of male characteristics (spermatangia and spermatia production);bi^+, wild allele that possibly inhibits the expression of female characteristics in male gametophytes; *bi*, mutant allele that would allow the expression of female characteristics in male gametophytes; and $mt^f mt^m$, gene that allows tetrasporophytic characteristic in diploid specimens (tetrasporangia)

$p < 0.0006$), α_{ETR} ($rd = 0.044 \pm 0.00$ and $gb = 0.058 \pm 0.00$; $F = 465.87$, $p < 0.000001$), and $\Delta F/F_m'$ ($rd = 0.575 \pm 0.01$ and $gb = 0.604 \pm 0.00$; $F = 27.63$, $p < 0.002$) when compared with red female gametophytes. However, red female gametophytes showed higher values of EK_{ETR} (45.00 ± 1.79) in relation to greenish-brown female gametophytes (38.35 ± 2.59; $F = 17.71$, $p > 0.005$).

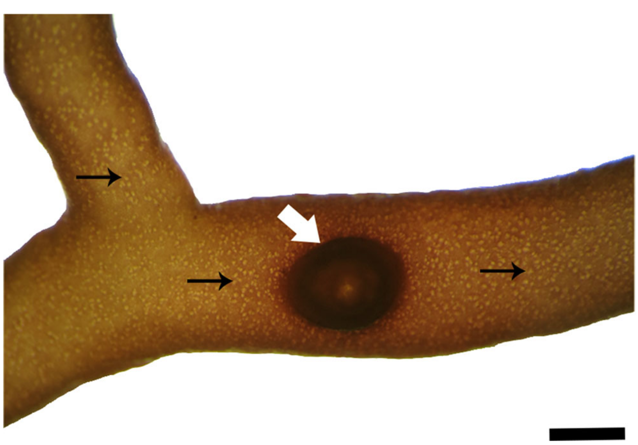

Fig. 4 Bisexual gametophytic thallus of *Gracilaria caudata* obtained from the germination of tetraspore released by *gbgb* tetrasporophyte. Black arrows point to spermatial crypts, and white arrow points to a cystocarp. Scale 1 mm

Attachment percentage Red female gametophytes showed higher attachment percentage (68.75 ± 23.93) on the flasks when compared with greenish-brown female gametophytes (25.0 ± 0.00) after a period of 14 days ($F = 8.00$, $p < 0.03$) (Fig. 5b).

Number of apices

Red and greenish-brown female gametophytes (*rd* and *gb* genotypes, respectively) showed differences in the number of apices, both when considering the whole experimental period of 21 days (41.5 ± 6.35 and 16.0 ± 5.47, respectively; $F = 36.98$, $p < 0.0008$) and over the 3 weeks ($F = 17.99$, $p < 0.0002$). For the first week of cultivation, no difference was observed ($p > 0.68$); however, the number of apices in red gametophytes increased during the second week (5.50 to 18.0; $p < 0.004$), while these numbers remained stable in

Table 2 *Gracilaria caudata*—offspring of carpospores obtained from cystocarps produced by seven bisexual gametophytes (*bi*) and their differentiated reproductive structures

Tetrasporophyte	Bisexual gametophytes	Offspring	Reproductive structures
gbgb	*bi* 02	Tetrasporophytes	Tetrasporangia
rdgb	*bi* 03	Male gametophytes	Spermatangia
rdgb	*bi* 04	Male gametophytes	Spermatangia
rdgb	*bi* 05	Male gametophytes	Spermatangia
gbrd	*bi* 06	Tetrasporophytes	Tetrasporangia
gbrd	*bi* 07	Tetrasporophytes	Tetrasporangia, spermatangia and one cystocarp
gbrd	*bi* 08	Male gametophytes	Spermatangia and one cystocarp

Parental tetrasporophytes of the seven bisexual gametophytes are shown in the first column

relation to the previous week in greenish-brown gametophytes (4.25 to 7.25; $p > 0.56$). In the third week, red female gametophytes had a higher number of apices in relation to the second week (18.0 to 41.5; $p < 0.0001$), as well as in relation to greenish-brown gametophytes for the third week ($p < 0.0001$). The latter showed an increase in the number of apices from the second to third week (7.25 to 16.0; $p < 0.01$) (Fig. 5c). Qualitatively, we also observed that red apices could be easily detached from the main thallus when compared with greenish-brown plants, giving origin to many detached fragments, which explains the need for 21 days of experiment.

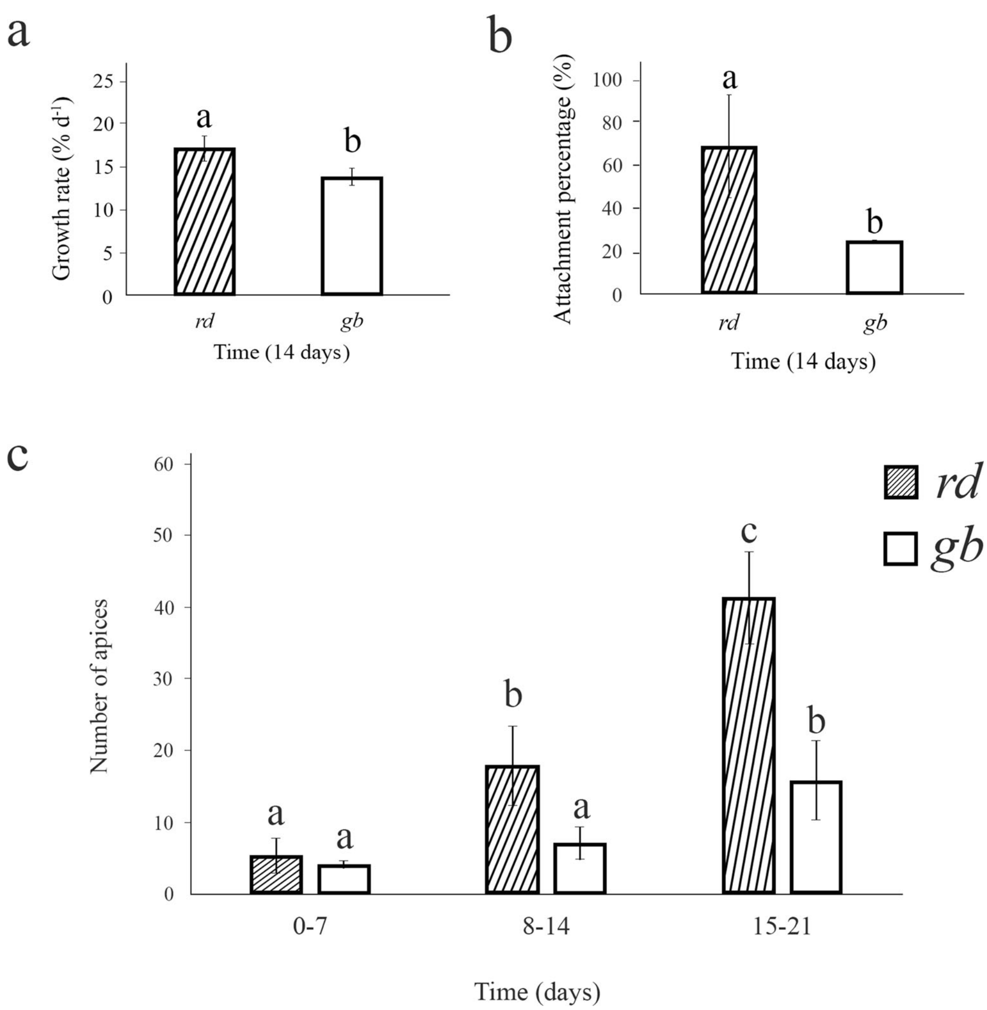

Fig. 5 *Gracilaria caudata*. (a) Growth rates of red (*rd*) and greenish-brown (*gb*) female gametophytes of *Gracilaria caudata* calculated from the average of fresh mass (mg) obtained after a period of 14 days under general cultivation conditions. (b) Attachment percentage of red (*rd*) and greenish-brown (*gb*) female gametophytes obtained after a period of 14 days of cultivation in irradiance of 25 ± 5 μmol photons $m^{-2}.s^{-1}$ without aeration and medium exchange. (c) Number of apices differentiated in red (*rd*) and greenish-brown (*gb*) female gametophytes during a period of 21 days under general cultivation conditions. Data are presented as average, and bars indicate standard deviation ($n = 4$). Different letters indicate significant differences according to one-way ANOVA and Newman-Keuls test ($p < 0.05$) for a and b and, rmANOVA and Newman-Keuls test ($p < 0.05$) for c

Discussion

The greenish-brown phenotype found in specimens from a natural population of *Gracilaria caudata* is stable (not reversible under general culture conditions) and the results of crosses performed in laboratory suggest that this phenotype is codominant in relation to the red phenotype and it comes from a nuclear mutation, refuting our initial hypothesis. In addition, analysis in the progeny of greenish-brown specimens highlight the presence of gametophytes with bisexual phenotype (monoicous), which is an atypical condition in the life cycle of Gracilariales. Greenish-brown female gametophytes showed a higher photosynthetic performance in culture when compared with the red female gametophytes. Even so, the latter still had the best performance in culture (growth rates, number of apices, and number of cystocarps), which coincides with its predominance in population.

Homozygous tetrasporophytes of *G. caudata* were red (*rdrd* genotype) or greenish-brown (*gbgb* genotype) and these phenotypes were transmitted to all individuals of their respective gametophytic progeny. However, heterozygous tetrasporophytes (*rdgb* and *gbrd* genotypes) presented the brown phenotype with parental color segregation in the gametophytic progeny in a ratio of 1:1 (Table 1). Both red and greenish-brown colors were codominant. This pattern of color nuclear inheritance has been poorly reported in *Gracilaria* (*G. tikvahiae*, van der Meer 1979a; and *G. domingensis*, Plastino et al. 1999) and also in Rhodophyta (van der Meer 1990). Most color mutants in Rhodophyta had a recessive nuclear inheritance (van der Meer 1990, for a review, Plastino et al. 2004, Costa and Plastino 2011), followed by cytoplasmic inheritance (van der Meer 1990).

Carpospores and tetraspores of *G. caudata* showed a red color, regardless of their origin. The greenish-brown and brown colors were observed only when the plants already had an erect axis (more than 2 cm and 4 cm in length, respectively). Differences of color between mutants and wild-type plants were commonly observed only in older plants, as in the green spontaneous mutant of *G. tikvahiae* (van der Meer and Bird 1977), *G. domingensis* (Plastino et al. 1999), *G. cornea* (Ferreira et al. 2006), *G. birdiae* (Costa and Plastino 2011), and in the greenish-brown individuals of *G. birdiae* (Costa and Plastino 2011). It is possible that cell and vacuole sizes in relation to the number of plastids determine the importance of color in young plants (Plastino et al. 1999). However, despite being poorly reported, spores and plantlets of spontaneous color mutants can also express color from the early stages of development, as observed for a light green mutant of *G. birdiae* (Plastino et al. 2004).

Regardless of the color phenotype (red or greenish-brown), the life cycle of *G. caudata* was completed in about 5 months, and it was one of the shortest times reported for *Gracilaria* species, which generally varies between 5 and 16 months (Ogata et al. 1972; Bird et al. 1977; McLachlan and Edelstein 1977; Bird et al. 1982, 1986; Oliveira and Plastino 1984; Rueness et al. 1987; Plastino and Oliveira 1988; Guimarães et al. 1999; Costa and Plastino 2001). This highlights the importance of this species as a possible model organism for reproductive studies of red algae. It is worth pointing out that the greenish-brown mutant of *G. caudata* completed its life cycle in a shorter period (145 days) in relation to the wild type (157 days). In addition, we observed a shorter time to complete the life cycle from crosses between gametophytes with different genotypes (*rd* × *gb*, 132 days; *gb* × *rd*, 138 days) since female fertilized plants from *rdgb* and *gbrd* genotypic tetrasporophytes developed cystocarps earlier (49 days of age), when compared with female plants from *gbgb* (56 days) and *rdrd* tetrasporophytes (62 days). The opposite was observed for other species in which the life cycle of the wild-type strain was completed in a shorter period (i.e., *G. birdiae*, Costa and Plastino 2001).

Despite the delay in the sexual differentiation of red female plants derived from tetraspores released by *rdrd* genotypic tetrasporophytes, they produced twice as many cystocarps per individual (39) when compared with the female fertilized gametophytes obtained from *gbgb*, *rdgb*, and *gbrd* genotypic tetrasporophytes (19, 18, and 19, respectively). The higher number of cystocarps observed in red female gametophytes suggests a greater amount of *rdrd* genotypic carpospores available to recruit in nature when compared with carpospores of other genotypes *gbgb*, *rdgb*, and *gbrd*, thus contributing to the predominance of wild-type specimens.

From red tetrasporophytes (*rdrd* genotype), we observed a 1:1 sexual ratio between female and male gametophytes, which is expected for *Gracilaria* species (Kain and Destombe 1995) and it is in agreement of the UV system of sex determination (Bachtrog et al. 2011). Only in the progeny of color mutant tetrasporophytes of *G. caudata* (*gbgb*, *rdgb*, and *gbrd* genotypes), the normal dioecious condition was disturbed owing to the presence of bisexual gametophytes (*bi*). This reproductive mutant spontaneously obtained in the laboratory was characterized by cystocarps on a thallus initially considered male by the presence of spermatangia. Considering the occurrence of cystocarps and spermatangia completely intermingled on the thallus, we suggest that such mutant could not have been originated from the coalescence of tetraspores during germination since the tetraspores were isolated (i.e., not grown in close vicinity). Besides that, from the coalescence of tetraspores, we could also expect the formation of several erect axes on a single primary attachment disc, and subsequently, each erect apex would become a branch from which only a single type of reproductive structure could be observed. Furthermore, the formation of these different reproductive structures on the same thallus could also not be due to failure during tetrasporangium cytokinesis that would promote non-disjunction of tetraspores after meiosis (van der Meer et al. 1984; van der Meer 1986; Plastino et al. 1999). The bisexual phenotype observed in *G. caudata* is a mutation. A similar reproductive mutant was previously reported for *G. tikvahiae*, and it was explained by the presence of a recessive allele known as *bi*. This allele would be distinct from the sex-determining *mt* locus, and it allows the expression of female functions in male gametophytes (van der Meer et al. 1984; van der Meer 1986). Therefore, the genotype of a normal male gametophyte of *G. caudata* would be mt^m/bi^+, whereas bisexual gametophytes would be mt^m/bi.

Since bisexuality is inheritable and no bisexual gametophytes were observed in the progeny of *rdrd* tetrasporophyte of *G. caudata*, we suppose that no red male or female gametophytes, which were initially used in the crosses, had the *bi* allele, so their genotypes were described as mt^m/bi^+ and mt^f/bi^+, respectively. In the progeny of *gbgb* tetrasporophyte, only female and bisexual gametophytes were observed in a ratio of 1:1. Because no exclusively male gametophytes were observed, we suggest that *gbgb* tetrasporophyte came from the cross between two reproductive mutants ($mt^f/bi \times mt^m/bi$) and would thus be heterozygous for *mt* and homozygous for *bi* locus with their genotype described as mt^f/mt^m, bi/bi. Therefore, from meiosis in tetrasporophytic generation, only two reproductive classes could be obtained in a ratio of 1:1, such as female gametophytes carrying the *bi* allele and bisexual gametophytes. From *rdgb* tetrasporophyte we observed female, male, and bisexual gametophytes with red or greenish-brown phenotype (Table 1). This result indicates that possibly *rdgb* genotypic tetrasporophyte comes from the cross between a red female gametophyte and a greenish-brown bisexual plant ($mt^f/bi^+ \times mt^m/bi$); thus it would be heterozygous for *mt* and *bi* loci, and its genotype could be described as mt^f/mt^m, bi^+/bi. Consequently, four reproductive classes would be expected in a ratio of 1:1:1:1 after the meiotic process in this tetrasporophyte, i.e., female (mt^f/bi^+), female carrying the *bi* allele (mt^f/bi), male (mt^m/bi^+), and bisexual (mt^m/bi). This supposed segregation of two independent genes was previously reported for *G. tikvahiae* (van der Meer et al. 1984; van der Meer 1986). The authors also suggest that the *bi* allele could not be phenotypically detected in female plants, which naturally express female characteristics (cystocarps, when fertilized), thus resulting in a proportion of two female gametophytes, one male, and one bisexual (2:1:1). In the progeny of *gbrd* genotypic tetrasporophyte of *G. caudata*, we also observed female, male, and bisexual gametophytes with red or greenish-brown color. This result corroborates the idea that greenish-brown female plants used initially in the crosses carried the *bi* allele, since no bisexual gametophytes were observed in the progeny of red male gametophyte crossed with red female gametophyte. Therefore, it is possible that greenish-brown female gametophytes would have the mt^f/bi genotype, while *gbrd* genotypic tetrasporophytes would have the mt^f/mt^m, bi^+/bi genotype, being also heterozygous for the *mt* and *bi* loci. Nevertheless, since red bisexual gametophytes and greenish-brown male gametophytes were found in the progeny of *rdgb* and *gbrd* genotypic tetrasporophytes, we could infer that bisexuality is not linked to color, that is, bisexuality is independently inherited from color.

In the progeny of bisexual gametophytes 03, 04, 05, and 08 of *G. caudata*, we found only male individuals. Four events could have triggered such results considering that bisexuality is expressed late in the development of an individual, and it has a recessive nuclear inheritance (van der Meer et al. 1984; van der Meer 1986): (i) self-fertilization; (ii) cross-fertilization with a second bisexual gametophyte, resulting in a diploid bisexual progeny (mt^m/mt^m, bi/bi); (iii) parthenogenesis, resulting in haploid bisexual progeny (mt^m/bi); and (iv) cross-fertilization with a normal male gametophyte mt^m/bi^+, giving rise to a diploid male progeny (mt^m/mt^m, bi/bi^+). Thus, the possibility remains that the male progeny of some of these bisexual gametophytes (03, 04, 05, and 08) would not have reached full reproductive maturity. In fact, in the male progeny from *bi* 08, we observed a single specimen that developed a cystocarp, which could corroborate such proposition (Table 2). Similar results were observed for *G. tikvahiae* (van der Meer et al. 1984; van der Meer 1986). In the progeny of bisexual gametophytes 02, 06, and 07, we observed only individuals with tetrasporangia. In this case, these bisexual individuals could be considered true monoecious since the released carpospores developed into tetrasporophytes completing the life cycle. Similar observations of *G. tikvahiae* van der Meer et al. (1984) proposed that such bisexual gametophytes would have complete femininity. We would like to emphasize that more studies would be necessary to confirm or refute the proposals related to sexual determination presented in this work. In this sense, the use of molecular markers linked to sex could bring great contributions to the understanding of sexual determination in *Gracilaria*.

Red female gametophytes of *G. caudata* showed higher growth rates when compared with the greenish-brown gametophytes, as previously observed for red and greenish-brown parental tetrasporophytes of *G. caudata* (Faria and Plastino 2016). This finding agrees with most studies on color mutants, either in the field for *G. debilis* and *G. edulis* (Veeragurunathan et al. 2016), or in the laboratory for *G. tikvahiae* (van der Meer 1979b), *G. domingensis* (Guimarães 2000), *G. birdiae* (Plastino et al. 2004), and *Ahnfeltia plicata* (Mansilla et al. 2014). It is true that similar higher growth rates have also been reported for *Eucheuma denticulatum* and *Kappaphycus striatus* as *Kappaphycus striatum* Gerung and Ohno 1997). Moreover, similar growth rates were observed for *Kappaphycus alvarezii* (Muñoz et al. 2004), *G. cornea* (Ferreira et al. 2006), *G. birdiae* (Ursi et al. 2003, 2013), *Hypneia pseudomusciformis* (as *H. musciformis* Yokoya et al. 2007), and *Mazzaella laminarioides* (Navarro 2015).

Despite the better photosynthetic performance of greenish-brown gametophytes of *G. caudata*, they showed lower growth than red gametophytes. Such mutant showed a lower concentration of phycoerythrin in relation to the red type (Faria and Plastino 2016), a pigment that besides being important in the photosynthetic process, it can be used by algae as a nitrogen reserve (García-Sánchez et al. 1993) and for photoprotection (Sinha et al. 1995; Figueroa et al. 2008). Therefore, it is possible that higher concentrations of phycoerythrin have contributed to better performance expressed by red specimens. In contrast, lower concentrations of phycoerythrin would possibly result in a smaller phycobilisome size, thus reducing the mechanical barrier caused by this pigment, which would favor higher irradiance reaching the reaction centers of PSII and, consequently, induce a greater excitation of chlorophyll a molecules, leading to an increase in electron transport rates.

A greater number of apices observed for red gametophytes of *G. caudata* when compared with greenish-brown gametophytes resulted from the higher growth rates. Because of the many branches, wild-type plants would have a greater chance of thallus fragmentation, which could later act as propagules favoring the dispersal of the species. However, these fragments could only perform this function if they had the ability to reattach to the substrate and persist in the environment, contributing to the increase in recruitment rates. Under laboratory conditions, apices of *G. caudata*, regardless of thallus color, could form a secondary attachment disc (SAD) and reattach to the substrate, which is not commonly reported for other *Gracilaria* species. After their excision from the main thallus, detached fragments of *G. domingensis* were unable to reattach to the substrate in laboratory conditions since they developed a callus at the basal part, which had no adhesion capacity (Ramlov et al. 2013). *G. chilensis* can recruit in soft bottom areas by sediment burial, but they are unable to form new holdfasts and thus cannot reattach to hard substrata (Guillemin et al. 2008). In *G. vermiculophylla*, despite the formation of new holdfast, no reattachment was observed in laboratory conditions (Nyberg and Wallentinus 2009). Furthermore, the persistence of its detached fragments in the environment has been reported as free-floating thallus that lacks holdfasts (Krueger-Hadfield et al. 2016).

Thallus fragmentation and secondary attachment discs observed in *G. caudata* might represent a vegetative propagation strategy, which would be more effective in specimens of red color when compared with those of greenish-brown color since the latter presented a lower number of apices differentiated and reattachment percentage. Moreover, this capacity could be one of the mechanisms contributing to the gene flow in *G. caudata* evidenced by Ayres-Ostrock et al. (2019) between populations located on the northeastern coast of Brazil. Secondary attachment discs have been frequently reported for Gigartinales species like *Chondracanthus squarrulosus* (Pacheco-Ruiz et al. 2005), *C. chamissoi* (González et al. 1997; Bulboa et al. 2005, 2013; Sáez et al. 2008; Sáez and Macchiavello 2018), *Eucheuma isiforme* and *E. nudum* (Dawes et al. 1974), *Gelidium rex* (Rojas et al. 1996), *G. sesquipedale* (Juanes and Puente 1993), and *Solieria filiformis* (Perrone and Cecere 1997). In these species, secondary attachment discs can be formed along the entire surface of the fragment in a manner different from what we observed for *G. caudata*, which forms its secondary attachment discs only in the basal part of the thallus. The reattachment of Gigartinales species occurs within 14 days of thallus contact with substrate, as observed to *G. caudata*. This capacity could be used for the recolonization of devastated areas by overharvesting (Dawes et al. 1974), reducing the time required for the implantation of new individuals in nature when compared with spore germination on artificial substrates for later placement in nature, as proposed by Miranda (2010). Furthermore, considering mariculture purposes, this secondary attachment capacity represents a maximization of the number of propagules, which could be obtained from a donor plant, thus representing a more sustainable alternative than a simple collection of biomass from natural banks, as commonly occurs on the Brazilian coast.

Conclusions

In the present study we verified a nuclear codominant inheritance of the greenish-brown color found in a natural mutant of *Gracilaria caudata*, which is poorly reported in red algae. Regardless of the color strain (red or greenish-brown), the life cycle was completed in 5 months, one of the shortest times reported for *Gracilaria*, emphasizing the importance of the species as a possible model organism for reproductive studies of red algae. Despite the lower fitness, greenish-brown gametophytes reached early reproductive maturity, as well as presented better photosynthesizing performance, contributing to its maintenance in the natural population. Differences found between mutant and red specimens contribute to the diversity and maintenance of the species, especially when considering its heterogeneous environment of occurrence (intertidal zone). Laboratory conditions differ from the natural environment where there are many biotic and abiotic factors interacting synergistically; thus physiological differences observed in the laboratory could be accentuated or minimized. Therefore, it would also be interesting to analyze agar yield and fitness of color mutants in experimental culture at sea, as well as their frequency in the natural population. Secondary attachment capacity was, for the first time, reported for a *Gracilaria* species, and it represents an alternative to spore reproduction, gene flow between populations, recovery of natural populations, and a new methodology to improve species mariculture. Bisexual gametophytes (reproductive mutant) were for the first time reported for *G. caudata*. We suggest the occurrence of the *bi* allele conferring bisexuality, which would be present in the greenish-brown tetrasporophyte from the field and transmitted to subsequent generations. Furthermore, the possibility remains that this *bi* allele is also present in red specimens in natural populations since we observed red *bi* gametophytes in the progenie of *rdgb* and *gbrd* tetrasporophytes.

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References

Ahmed S, Cock JM, Pessia E, Luthringer R, Cormier A, Robuchon M (2014) A haploid system of sex determination in the brown alga *Ectocarpus* sp. Curr Biol 24:1945–1957

Araújo FO, Ursi S, Plastino EM (2014) Intraspecific variation in *Gracilaria caudata* (Gracilariales, Rhodophyta): growth, pigment content, and photosynthesis. J Appl Phycol 26:849–858

Ayres-Ostrock LM, Mauger S, Plastino EM, Oliveira MC, Valero M, Destombe C (2016) Development and characterization of microsatellite markers in two agarophyte species, *Gracilaria birdiae* and *Gracilaria caudata* (Gracilariaceae, Rhodophyta), using next generation sequencing. J Appl Phycol 28:653–662

Ayres-Ostrock LM, Valero M, Mauger S, Oliveira MC, Plastino EM, Guillemin ML, Destombe C (2019) Dual influence of terrestrial and marine historical processes on the phylogeography of the Brazilian intertidal red alga *Gracilaria caudata*. J Phycol 55: 1096–1114

Bachtrog D, Kirkpatrick M, Mank JE, McDaniel DF, Pires JC, Rice WR, Valenzuela N (2011) Are all sex chromosomes created equal? Trends Genet 27:350–357

Bird NL, McLachlan J, Grund D (1977) Studies on *Gracilaria*. 5 In vitro life history of *Gracilaria* sp. from the maritime provinces. Can J Bot 55:1282–1290

Bird CJ, van der Meer JP, McLachlan J (1982) A comment on *Gracilaria verrucosa* (Huds.) Papenf. (Rhodophyta: Gigartinales). J Mar Biol 62:453–459

Bird CJ, McLachlan J, Oliveira EC (1986) *Gracilaria chilensis* sp. nov. (Rhodophyta, Gigartinales), from Pacific South America. Can J Bot 64:2928–2934

Bulboa C, Macchiavello J, Oliveira E, Fonck E (2005) First attempt to cultivate the carrageenan producing seaweed *Chondracanthus chamissoi* (Rhodophyta; Gigartinales) in northern Chile. Aquac Res 36:1069–1074

Bulboa C, Véliz K, Sáez F, Sepúlveda C, Vega L, Macchiavello J (2013) A new method for cultivation of the carragenophyte and edible red seaweed *Chondracanthus chamissoi* based on secondary attachment disc: development in outdoor tanks. Aquaculture 410/411:86–94

Câmara-Neto (1987) Seaweed culture in Rio Grande do Norte, Brazil. Hydrobiologia 151/152:363–367

Chiaramonte RA, Parra AP, Ayres-Ostrock LM, Plastino EM (2019) *Gracilaria caudata* (Gracilariales, Rhodophyta) is reproductively compatible along the whole Brazilian coast. J Appl Phycol 31: 931–937

Coelho SM, Gueno J, Lipinska AP, Cock JM, Umem JG (2018) UV chromosomes and haploid sexual systems. Trends Plant Sci 23: 974–807

Coelho SM, Mignerot L, Cock JM (2019) Origin and evolution of sex-determination systems in the brown algae. New Phytol 222:1751–1756

Cohen S, Faugeron S, Martınez EA, Correa JA, Viard F, Destombe C, Valero M (2004) Molecular identification of two sibling species under the name *Gracilaria chilensis* (Rhodophyta, Gracilariales). J Phycol 40:742–747

Costa VL, Plastino EM (2001) Histórico de vida de espécimens selvagens e variantes cromáticas de *Gracilaria birdiae* (Gracilariales, Rhodophyta). Rev Bras Bot 24:491–500

Costa VL, Plastino EM (2011) Color inheritance and pigment characterization of red (wild-type), greenish-brown, and green strains of *Gracilaria birdiae* (Gracilariales, Rhodophyta). J Appl Phycol 23: 599–605

Costa ES, Plastino EM, Petti R, Oliveira EC, Oliveira MC (2012) The Gracilariaceae Germplasm Bank of the University of São Paulo, Brazil—a DNA barcoding approach. J Appl Phycol 24:1643–1653

Dawes CJ, Mathieson AC, Cheney DP (1974) Ecological studies of Floridian *Eucheuma* (Rhodophyta, Gigartinales). I. Seasonal growth and reproduction. Bull Mar Sci 24:235–273

Destombe C, Valero M, Guillemin ML (2010) Delineation of two sibling red algal species, *Gracilaria gracilis* and *Gracilaria dura* (Gracilariales, Rhodophyta), using multiple DNA markers: resurrection of the species *G. dura* previously described in the northern Atlantic 200 years ago. J Phycol 46:720–727

Faria AVF, Plastino EM (2016) Physiological assessment of the mariculture potential of a *Gracilaria caudata* (Gracilariales, Rhodophyta) variant. J Appl Phycol 28:2445–2452

Faria AVF, Barufi JB, Plastino EM (2017) Ecotypes of *Gracilaria caudata* (Gracilariales, Rhodophyta): physiological and morphological approaches considering life history phases. J Appl Phycol 29: 707–719

Ferreira LB, Barufi JB, Plastino EM (2006) Growth of red and green strains of the tropical agarophyte *Gracilaria cornea* J. Agardh (Gracilariales/Rhodophyta) in laboratory. Rev Bras Bot 29:187–192

Figueroa FL, Bueno A, Korbee N, Santos R, Mata L, Schuenhoff A (2008) Accumulation of mycosporine-like amino acids in *Asparagopsis armata* grown in tanks with fishpond effluents of gilthead sea bream, *Sparus aurata*. J World Aquacult Soc 39:692–699

García-Sánchez MJ, Fernández JA, Niell FX (1993) Biochemical and physiological responses of *Gracilaria tenuistipitata* under two different nitrogen treatments. Physiol Plant 88:631–637

Gerung GS, Ohno M (1997) Growth rates of *Eucheuma denticulatum* (Burman) Collins et Harvey and *Kappaphycus striatum* (Schmitz) Doty under different conditions in warm waters of southern Japan. J Appl Phycol 9:413–415

González J, Meneses I, Vasquez J (1997) Field studies in *Chondracanthus chamissoi* (C. Agardh) Kützing: seasonal and spatial variations in life-cycle phases. Biologia Pesquera 26:3–12

Guillemin ML, Faugeron S, Destombe C, Viard F, Correa JA, Valero M (2008) Genetic variation in wild and cultivated populations of the haploid– diploid red alga *Gracilaria chilensis*: how farming practices favor asexual reproduction and heterozygosity. Evolution 62: 1500–1519

Guillemin ML, Huanel OR, Martínez EA (2012) Characterization of genetic markers linked to sex determination in the haploid-diploid red alga *Gracilaria chilensis*. J Appl Phycol 48:365–372

Guimarães M (2000) Aspectos fisiológicos de *Gracilaria domingensis* (Gracilariales, Rhodophyta): subsídio para a compreensão da manutenção do polimorfismo pigmentar. Doctoral thesis, University of São Paulo.

Guimarães M, Plastino EM, Oliveira EC (1999) Life History, Reproduction and Growth of *Gracilaria domingensis* (Gracilariales, Rhodophyta) from Brazil. Bot Mar 42:481–486.

Guiry, M.D., Norris, J.N., Fredericq, S. & Gurgel C.F.D. (2018). *Crassiphycus* Guiry, Gurgel, J.N.Norris & Fredericq, gen. nov., a replacement name for *Crassa* Gurgel, J.N.Norris & Fredericq, nom. inval. (Gracilariaceae, Rhodophyta), with some additional nomenclatural notes. Notulae Algarum 82: 1–4

Gurgel CFD, Norris JN, Schmidt WE, Le HN, Fredericq S (2018) Systematics of the Gracilariales (Rhodophyta) including new subfamilies, tribes, subgenera, and two new genera, gen nov and *Crassa* gen nov, *Agarophyton*. Phytotaxa 374:1–23

Hayashi L, Bulboa C, Kradolfer P, Soriano G, Robledo D (2014) Cultivation of red seaweeds: a Latin American perspective. J Appl Phycol 26:719–727

Iha C, Grassa CJ, Lyra GM, Davis CC, Verbruggen H, Oliveira MC (2018) Organellar genomics: a useful tool to study evolutionary relationships and molecular evolution in Gracilariaceae (Rhodophyta). J Phycol 54:775–787

Juanes JA, Puente A (1993) Differential reattachment capacity of isomorphic life history phases of *Gelidium sesquipedale*. Hydrobiologia 260/261:139–144

Kain JM, Destombe C (1995) A review of the life history, reproduction and phenology of *Gracilaria*. J Appl Phycol 7:269–281

Krueger-Hadfield SA, Kollars NM, Byers JE, Greig TW, Hammann M, Murray DC, Murren CJ, Strand AE, Terada R, Weinberger F, Sotka EE (2016) Invasion of novel habitats uncouples haplodiplontic life cycles. Mol Ecol 25:3801–3816

Levy I, Friedlander M (1990) Strain selection in *Gracilaria* spp. I. Growth, pigment and carbohydrates characterization of strains of *G. conferta* and *G. verrucosa* (Rhodophyta, Gigartinales). Bot Mar 33:339–345

Lyra GDM, Gurgel CFD, Costa EDS, Jesus PB, Oliveira MC, Oliveira EC, Davis CC, Nunes JMDC (2016) Delimitating cryptic species in the *Gracilaria domingensis* complex (Gracilariaceae, Rhodophyta) using molecular and morphological data. J Phycol 52:997–1017

Mansilla A, Rodriguez JP, Souza JMC, Rosenfeld S, Ojeda J, Yokoya NS (2014) Growth responses to temperature, salinity and nutrient variations, and biomass variation and phenology of *Ahnfeltia plicata* (Rhodophyta, Ahnfeltiales): a commercially interesting agarophyte from the Magellanic region, Chile. J Appl Phycol 26:1133–1139

Martinez EA, Destombe C, Quillet MC, Valero M (1999) Identification of random amplified polymorphic DNA (RAPD) markers highly linked to sex determination in the red alga *Gracilaria gracilis*. Mol Ecol 8:1533–1538

McHugh DJ (2003) A guide to the seaweed industry. In FAO Fisheries Techinal Paper. Food and Agriculture Organization of the United Nations. Rome, pp 441

McLachlan J, Edelstein T (1977) Life history and culture of *Gracilaria foliifera* (Rhodophyta) from South Devon. J Mar Biol 57:577–586

Miranda G (2010) Monitoramento, manejo e restauração de populações naturais de *Gracilaria caudata* J. Agardh (Rhodophyta, Gracilariales) subsídios ao uso sustentável. Doctoral thesis, University of São Paulo

Miranda GEC, Yokoya NS, Fujii MT (2012) Effects of temperature, salinity and irradiance on carposporeling development of *Hidropuntia caudata* (Gracilariales, Rhodophyta). Braz J Pharmacog 22:818–824

Muñoz J, Freile-Pelegrín Y, Robledo D (2004) Mariculture of *Kappaphyccus alvarezii* (Rhodophyta, Solieriaceae) color strains in tropical waters of Yucatán, México. Aquiculture 239:161–177

Navarro P (2015) Presencia de variante verde de *Mazzaella laminarioides* (Gigartinales, Rhodophyta) en la Región de Magallanes, Chile. Rev Biol Mar Oceanogr 50:193–198

Nyberg CD, Wallentinus I (2009) Long-term survival of an introduced red alga in adverse conditions. Mar Biol Res 5:304–308

Ogata E, Matsui T, Nakamura H (1972) The life cycle of *Gracilaria verrucosa* (Rhodophyceae, Gigartinales) in vitro. Phycologia 11: 75–80

Oliveira EC, Miranda GEC (1998) Aspectos sociais e econômicos da explotação de algas marinhas do Brasil. In: Paula EJ, Cordeiro-Marino M, Santos DP, Plastino EM, Fujii MT, Yokoya NS (eds) Anais IV Congresso Latino Americano, II Reunião Libero-Americana e VII Reunião Brasileira de Ficologia. Sociedade Ficológica da América Latina e Caribe, Sociedade Brasileira de Ficologia, São Paulo, pp. 149–156

Oliveira EC, Plastino EM (1984) The life-history of *Gracilaria* (Rhodophyta) from Brazil. Jap J Phycol 32:203–208

Pacheco-Ruiz I, Zertuche J, Espinoza J (2005) The role of secondary attachment disc in the survival of *Chondracanthus squarrulosus* (Rhodophyta, Gigartinales). Phycologia 44:629–631

Perrone C, Cecere E (1997) Regeneration and mechanisms of secondary attachment in *Solieria filiformis* (Gigartinales, Rhodophyta). Phycologia 36:120–127

Plastino EM, Oliveira EC (1988) Deviations in the life-history of *Gracilaria* sp. (Rhodophyta, Gigartinales), from Coquimbo, Chile, under different culture conditions. Hydrobiologia 164:67–74

Plastino EM, Oliveira EC (1990) Crossing experiments as an aid to the taxonomic recognition of the agarophytes Gracilaria. In: Kautsky N, Oliveira EC (eds) Cultivation of seaweeds in Latin America, São Paulo, pp 127–133

Plastino EM, Oliveira EC (1997) *Gracilaria caudata* J. Agardh (Gracilariales, Rhodophyta) - restoring an old name for a common western Atlantic alga. Phycologia 36:225–232

Plastino EM, Guimarães M, Matioli SR, Oliveira EC (1999) Codominant inheritance of polymorphic color variants of *Gracilaria domingensis* (Gracilariales, Rhodophyta). Genet Mol Biol 22:105–108

Plastino EM, Ursi S, Fujii MT (2004) Color inheritance, pigment characterization, and growth of a rare light green strain of *Gracilaria birdiae* (Gracilariales, Rhodophyta). Phycol Res 52:45–52

Platt T, Gallegos CL, Harrison WG (1980) Photoinhibition of photosynthesis in natural assemblages of marine phytoplankton. J Mar Res 38:687–701

Porse H, Rudolf B (2017) The seaweed hydrocolloid industry: 2016 updates, requirements, and outlook. J Appl Phycol 29:2187–2200

Ramlov F, Plastino EM, Yokoya N (2013) Growth, callus formation and plant regeneration in color morphs of *Gracilaria domingensis* (Gracilariales, Rhodophyta) cultured under different irradiance and plant growth regulators. Phycologia 52:508–516

Robinson N, Winberg P, Kirkendale L (2013) Genetic improvement of macroalgae: status to date and needs for the future. J Appl Phycol 25:703–716

Rojas RH, Léon NM, Rojas RO (1996) Practical and descriptive techniques for *Gelidium rex* (Gelidiales, Rhodophyta) culture. Hydrobiologia 326:367–370

Rueness J, Mathisen HA, Tananger T (1987) Culture and field observations on *Gracilaria verrucosa* (Huds.) Papenf. (Rhodophyta) from Norway. Bot Mar 30:267–276

Sáez F, Macchiavello F (2018) Secondary attachment discs: a new alternative for restoring populations of *Chondracanthus chamissoi* (Gigartinales, Rhodophyta). Lat Am J Aquat Res 46:140–146

Sáez F, Macchiavello J, Fonck E, Bulboa C (2008) The role of the secondary attachment disc in the vegetative propagation of *Chondracanthus chamissoi* (Gigartinales; Rhodophyta). Aquat Bot 89:63–65

Schreiber U (2004) Pulse-amplitude-modulation (PAM) fluorometry and saturation pulse method: an overview. Papageorgiou GC, Govindjee (eds) Chlorophyll *a* fluorescence: a signature of photosynthesis. Springer, Dordrecht, pp 279–319

Sim MC, Lim PE, Gan SY, Phang SM (2007) Identification of random amplified polymorphic DNA (RAPD) marker for differentiating male from female and sporophytic thalli of *Gracilaria changii* (Rhodophyta). J Appl Phycol 19:763–769

Simioni C, Hayashi L, Oliveira MC (2019) Seaweed resources of Brazil: what has changed in 20 years? Bot Mar 62:433–441

Sinha RP, Lebert M, Kumar A, Kumar HD, Häder DP (1995) Spectroscopic and biochemical analyses of UV effects on phycobilisomes of *Anabaena* sp. and *Nostoc carmium*. Bot Acta 108:87–92

Torres P, Santos JP, Chow F, dos Santos DYAC (2019) A comprehensive review of traditional uses, bioactivity potential, and chemical diversity of the genus *Gracilaria* (Gracilariales, Rhodophyta). Algal Research 37:288–306.

Ursi S, Plastino EM (2001) Crescimento in vitro de linhagens de coloração vermelha e marrom-esverdeada clara de *Gracilaria* sp. (Gracilariales, Rhodophyta) em dois meios de cultura: análise de diferentes estádios reprodutivos. Rev Bras Bot 24:587–594

Ursi S, Perdesén M, Plastino E, Snoeijs P (2003) Intraspecific variation of photosynthesis, respiration and photoprotective carotenoids in *Gracilaria birdiae* (Gracilariales: Rhodophyta). Mar Biol 142: 997–1007

Ursi S, Costa VL, Hayashi L, Pereira RTL, Paula EJ, Plastino EM (2013) Intraspecific variation in *Gracilaria birdiae* (Gracilariales, Rhodophyta): growth, and agar yield and quality of color strains under aquaculture. Bot Mar 56:241–248

van der Meer JP (1977) Genetics of *Gracilaria* sp. (Rhodophyceae, Gigartinales). II. The life history and genetic implications of cytokinetic failure during tetraspore formation. Phycologia 16:367–371

van der Meer (1978) Genetics of *Gracilaria* sp. (Rhodophyceae, Gigartinales) III. Non-Mendelian gene transmission. Phycologia 17:314–318

van der Meer JP (1979a) Genetics of *Gracilaria* sp. (Rhodophyceae, Gigartinales). III. Non Mendelian gene transmission. Phycologia 17:314–318

van der Meer JP (1979b) Genetics of *Gracilaria* sp. (Rhodophyceae, Gigartinales). V. Isolation and characterization of mutant strains. Phycologia 18:47–54

van der Meer JP (1986) Genetics of *Gracilaria tikvahiae* (Rhodophyceae) XI Further characterization of a bisexual mutant. J Phycol 22:151–158

van der Meer JP (1987) Marine algal genetics and genomes. Hydrobiologia 151/152:49–56

van der Meer JP (1990) Genetics. In: Cole KM, Sheath RG (eds) Biology of the red algae. Cambridge University Press, Cambridge, pp 103–122

van der Meer JP, Bird NL (1977) Genetics of *Gracilaria* sp. (Rhodophyceae, Gigartinales). I. Mendelian inheritance of two spontaneous green variants. Phycologia 16:159–161

van der Meer JP, Tood ER (1977) Genetics of *Gracilaria* sp. (Rhodophyceae, Gigartinales). IV. Mitotic recombination and its relationship to mixed phases in the life history. Can J Bot 55: 2810–2817

van der Meer JP, Patwary MU, Bird CJ (1984) Genetics of *Gracilaria tikvahiae* (Rhodophyceae) X Studies on a bisexual clone. J Phycol 20:42–46

Veeragurunathan V, Prasad K, Singh N, Malarvizhi J, Mandal SK, Mantri VA (2016) Growth and biochemical characterization of green and red strains of the tropical agarophytes *Gracilaria debilis* and *Gracilaria edulis* (Gracilariaceae, Rhodophyta). J Appl Phycol 28:3479–3489

Yokoya NS, Necchi O, Martins AP, Gonzalez SF, Plastino EM (2007) Growth responses and photosynthetic characteristics of wild and phycoerythrin-deficient strains of *Hypnea musciformis* (Rhodophyta). Journal of Applied Phycology 19 (3):197–205

Yokoya NS, Oliveira EC (1992a) Effects of salinity on the growth rate, morphology and water content of some Brazilian red algae of economic importance. Sci Mar 18:49–64

Yokoya NS, Oliveira EC (1992b) Temperature response of economically important red algae and their potential for mariculture in Brazilian waters. J Appl Phycol 4:339–345

Yong YS, Yong WTL, Anton A (2013) Analysis of formulae for determination of seaweed growth rate. J Appl Phycol 25:1831–1834

Zar JH (1996) Biostatistical analysis. Prentice Hall, Upper Saddle River

Zhang X, van der Meer JP (1988) A genetic study on *Gracilaria sjoestedtii*. Can J Bot 66:2022–2026